



Testeo Automatizado

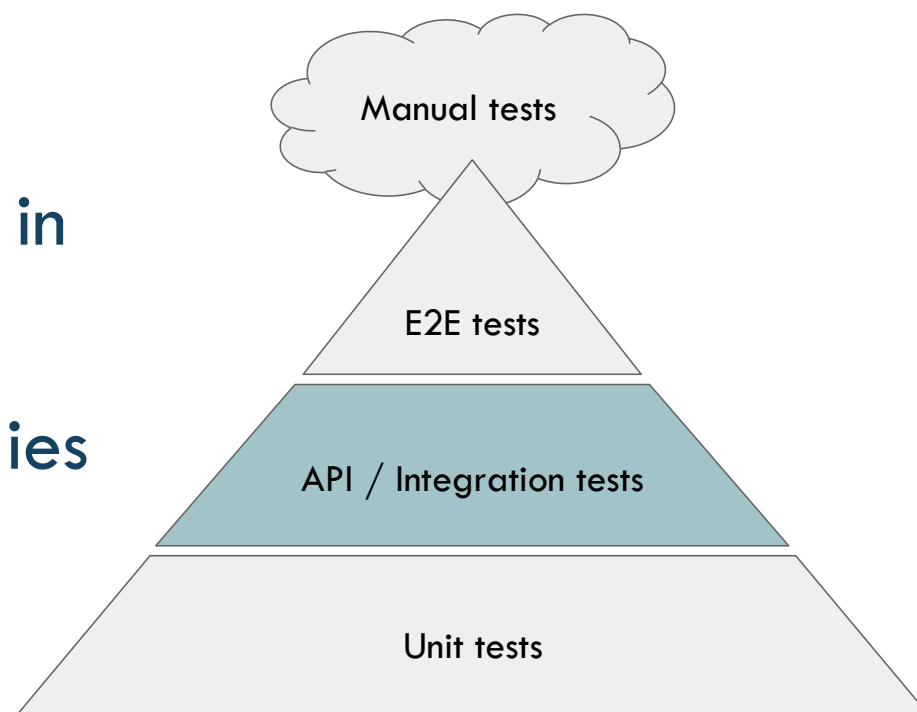
TESTEO AUTOMATIZADO

API / Integration Testing



AGENDA

- What API testing consists in
- Characteristics
- Patterns and methodologies
- Tools for API testing





WHAT IS API TESTING?

- Testing the **contract** between 2 systems
- For Web applications, generally testing REST / HTTP calls
- Either done by QAs or Developers



CHARACTERISTICS OF API TESTS

- Slower to execute than Unit tests
- More fragile
 - ↪ more underlying dependencies
- More difficult to setup
 - ↪ need some sessions
 - ↪ data setup in DB or previous API calls
 - ↪ Test data management (TDM) becomes an issue
- Tests interdependencies trap



AAA TESTING PATTERN

- AAA pattern still applies here
- A for **Arrange**
 - ↪ setup initial data (session, JWT, etc...)
- A for **Act**
 - ↪ Call endpoint under test
- A for **Assert**
 - ↪ Validate endpoint response (HTTP status code, response body, etc...)



WHAT ABOUT TDD?

- TDD can be applied too
- Example
 - ↪ QA creates API tests based on agreed contract
 - ↪ Test fails in CI/CD until developer submits code
 - ↪ Test succeeds if developer properly implemented contract



API TESTING TOOLS

- What you need
 - ↪ Easy to use by your QAs/Developers
 - ↪ Test maintenance easy
 - ↪ Fast to execute
 - ↪ Integratable with CI/CD pipelines
 - ↪ Easy to add more tests and scale
- Many tools respond to these criteria
 - ↪ Postman (most popular) for manual and automated tests
 - ↪ Mocha: programmatic, more Dev oriented
 - ↪ Almost any JS based testing framework would do the work
 - ↪ SoapUI
 - ↪ Others in any programming language



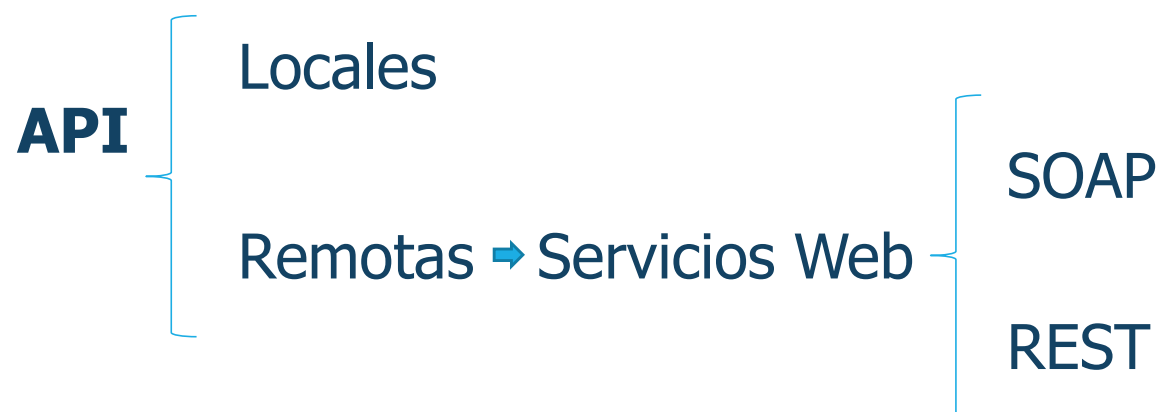


MOCKING FOR API TESTING

- Mock servers
 - ↪ control behavior of some dependencies
 - ↪ more robust tests
 - ↪ difficult to maintain
- Fake sessions and tokens
 - ↪ ability to programmatically generate session or JWTs



APPLICATION PROGRAMMING INTERFACE





REST

REST es una arquitectura para aplicaciones en redes (**RE**presentational **State Transfer**).

Roy Fielding



REST (REPRESENTATIONAL STATE TRANSFER)

- Interfaces estandarizadas
- Stateless
- Cacheable
- Sistema de capas
- Cliente / Servidor
- Código bajo demanda (opcional)



CÓDIGOS DE ESTADO

1xx - Respuestas informativas

2xx - Peticiones correctas

3xx - Redirecciones

4xx - Errores del cliente

5xx - Errores de servidor



PRACTICE

Configure **Postman** using environments so you have:

- Test environment
- Prod environment

Configurations should include at least

- base URL

Investigate

- how to run Postman tests in command line with **Newman**
- how to integrate Newman in CI/CD

<https://restful-booker.herokuapp.com/apidoc/index.html#api-Booking>



NEWMAN

Nos permite ejecutar y probar colecciones de Postman desde la línea de comandos.

Se pueden ejecutar desde un CI



- ***npm install -g newman***
- `newman run arch.postman_collection.json -e arch.postman_environment.json`
- ***npm install -g newman-reporter-htmlextra***
- `newman run arch.postman_collection.json -e arch.postman_environment.json --r htmlextra`



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MUCHAS GRACIAS